
Allen Huang

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Experience

Senior Software Engineer **Klaviyo** **Jul 2024 - Dec 2025**

- Built the asynchronous flow-recommendation pipeline, fanning work out across 5+ partner-domain microservices via Pulsar/Kafka; modeled generation as a finite-state machine in MySQL and cached expensive compute in Redis, reducing new-customer subsequent flow generations by 66%.
- Engineered the flow-classification backend that queues requests through Pulsar to the Data Science platform.
- Owned privacy/data-deletion compliance for the Flows domain: designed the cascading hard-delete pipeline that purges a customer's Flow object and 10+ dependent entities while preserving referential integrity across services.
- Participated in the on-call rotation for Flows; resolved P0/P1 incidents, debugged production issues in pairs, and built reusable CLI tooling that detected and repaired invalid state — eliminating a class of manual cleanup.
- Shipped the developer flow-update API and partnered with Design and Product on flow-graph path-analysis tooling; maintained supporting infrastructure on AWS with Terraform and Kubernetes.

Software Engineer **Klaviyo** **May 2022 - Jul 2024**

- Architected the "Encoded Flows" schema-transformation system to decouple internal storage from the public contract: redesigned internal relational models into rank-indexed FlowPaths/FlowActions (a DAG) and designed a separate serialized nodes-and-edges graph format that deliberately hides storage internals from the API. Owned the majority of the design and implementation in collaboration with my manager and the APIs team, with a comprehensive test suite — this became the foundation for the developer CRUD APIs, flow classification, and recommendation generation.
- Distributed transformation work across consumers via Pulsar so independent tasks could be processed asynchronously, decoupling throughput from single-threaded transformation logic.
- Built the core developer flow create/read APIs on top of the Encoded Flows format, now consumed by third-party integrations at 100+ RPS.
- Collaborated with Data Science on "FlowsAI" suggestions, combining LLM output with serialized flow graphs to generate recommended automations.
- Modernized Backbone.js analytics pages to React/Redux in partnership with Design/Product.

Software Engineer Intern **Klaviyo** **Jun 2021 - Aug 2021**

- Migrated Backbone.js analytics pages to React and built Django API endpoints for Cassandra data access, with pytest/Django test coverage for local and CI/CD execution.

Software Engineer Intern **PCI** **Sep 2021 - Dec 2021**

- Migrated the application from a GCP Kubernetes cluster to AWS and stood up a CI/CD pipeline (Jenkins, Trivy, Anchore, Harbor) with container image scanning.

Skills

- Languages:** Python, TypeScript, SQL, Bash, Nix, JavaScript
- Backend & Messaging:** Django, Kafka, Pulsar, RabbitMQ, Redis, MySQL, PostgreSQL, Clickhouse
- Distributed Systems:** event-driven architecture, async fan-out across services, finite-state-machine orchestration, decoupling internal data models from public API contracts
- Infrastructure & Cloud:** AWS, Kubernetes, Terraform, Docker, ArgoCD, Buildkite, CI/CD, Git
- Practices & Observability:** pair programming, code review, on-call/incident response, test-driven development (pytest, Playwright), Grafana, Sentry, Splunk, PagerDuty
- Languages:** English (native), Cantonese Chinese (conversational)

Education

B.S. in Computer Science, Oregon State University — Corvallis, OR **GPA: 3.95/4.0**

Awards & Patents

Multiple Category Wins, Klaviyo Hackathon **2024**

Third Place, Klaviyo Hackathon **2023**

Projects

Wikiquiz *Crystal, Svelte, Python, FastAPI, Redis, Terraform, Docker, spaCy, BERT, ONNX* **2021**